Automatic metadata-driven reanalysis in cloud infrastructures

ELIXIR All Hands 2023

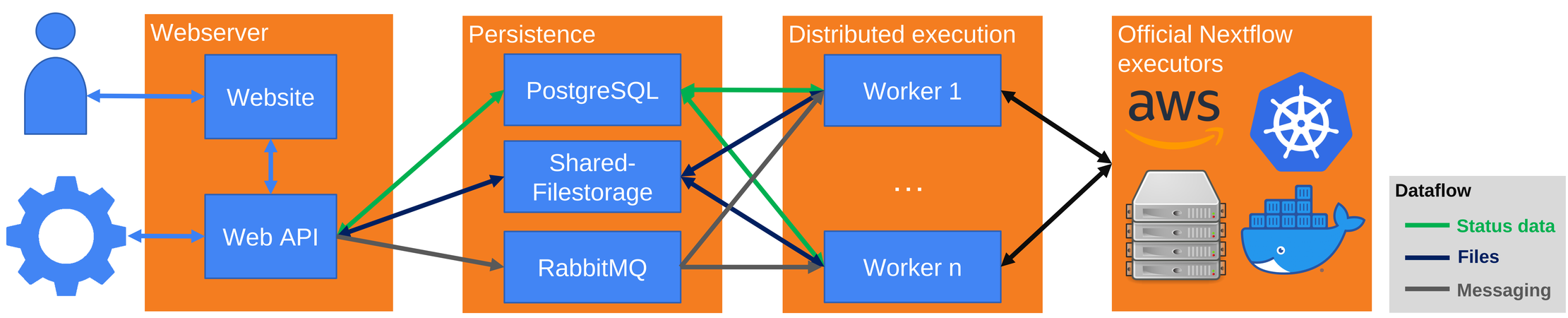
SDRF – Sample Data Reference Format

- Published in 2020 by Perez-Riverol et al.
- Automatic reanalysis depends heavily on well-annotated metadata metadata
- Build Nextflow workflows which use SDRF as input
- As SDRF is a relatively new the study aims to collect problems, ideas and impressions to help improving it
- Working on RDMKit content

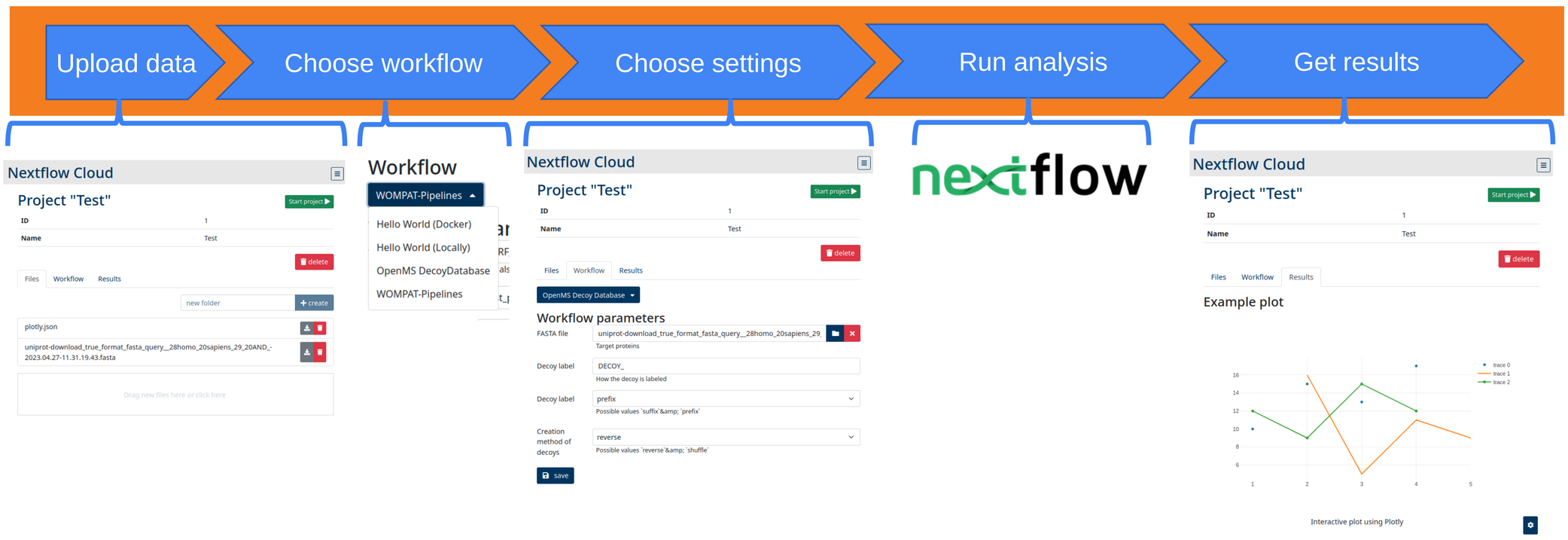
source name	characteristics[organism]	characteristics[organism part]	…
Name1	Homo Sapiens	Tissue	
Name2	E. Coli	Cell culture	

NF-Cloud

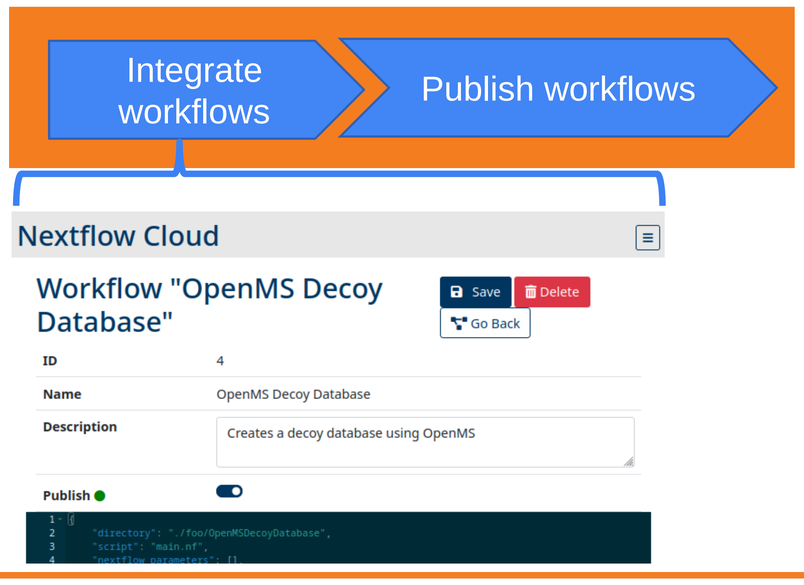
Architecture



For users



For developers



Advantages

- Graphical user interface for researchers
- OS independent
- Customizable/downloadable/interactive result plots
- Full support of Nextflow executors (local and/or cloud)
- Input validation
- Web-API to include or automate workflow execution
- Self hostable / Fully free
- FAIR

https://github.com/mpc-bioinformatics/nf-cloud

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